Production Theory for Computable General Equilibrium (CGE) Model

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Outline

- Production function
  - General relationship between inputs and output - *technology*
  - Returns to scale - the norm of CRS in CGE models
- A class of Constant Elasticity of Substitution (CES) Production function
  - The general CES production function
  - The Leontief production function
  - The linear production function
  - The Cobb-Douglas production function
- Producer's behaviour: *Cost Minimization*
  - First order conditions and demand for factors of production
    - General CES
    - Cobb-Douglas
    - Leontief
  - Illustration (with MathCAD) and comprehension test
- Nested production function

Production function

- A relationship between quantity of inputs and quantity of outputs
- Representing production technology for a specific firm or industry

$$Q = f\left(X_1, \ldots, X_i, \ldots, X_n\right)$$

- Q = quantity of output
  X_i = quantity of input i, $i = 1, \ldots, n$
- A well-behaved function has the following characteristics:

$$\frac{\partial Q}{\partial X_i} > 0 \text{ and } \frac{\partial^2 Q}{\partial X_i^2} < 0 \text{ for all } i = 1, \ldots, n$$

Production function (General-with technical change parameters)

- Adding standard technical change parameters:

$$Q = \frac{1}{A} f\left(\frac{X_1}{A_1}, \ldots, \frac{X_i}{A_i}, \ldots, \frac{X_n}{A_n}\right)$$

- A = All inputs-saving technical change (Total Factor Productivity)
  A_i = Factor-saving technical change
- $\tilde{X}_i = \frac{X_i}{A_i}$ is called effective input i.
- If A (TFP) increase/decrease, it means technical regress/progress
- If A_i increase/decrease, it means input-i-saving technical regress/progress

Homogeneity and returns to scale (CGE standard assumption)

- A function $f(\cdot)$ of n variables for which $(tX_1, ..., tX_n)$ is in the domain whenever $t > 0$ and $(X_1, ..., X_n)$ is in the domain is homogeneous of degree k if:

$$t^k f(X_1, \ldots, X_n) = f(tX_1, \ldots, tX_n)$$

- A production function $f(\cdot)$ is said to exhibit:
  - Increasing returns to scale if $k > 1$
  - Constant returns to scale if $k = 1$
  - Decreasing returns to scale if $k < 1$
- Example, the following production function:

$$f(X_1, X_2) = X_1^{\alpha} X_2^{1-\alpha}$$

exhibits constant returns to scale because:

$$\begin{aligned}(tX_1)^{\alpha}(tX_1)^{1-\alpha} &= t^{\alpha} X_1^{\alpha} t^{1-\alpha} X_2^{\alpha} = t^{\alpha} t^{1-\alpha} X_1^{\alpha} X_2^{\alpha} \\ &= t^{\alpha+1-\alpha} X_1^{\alpha} X_2^{\alpha} = t^1 f(X_1, X_2)\end{aligned}$$

Constant elasticity of substitution (CES) Production function

- General form of CES production fuction:

$$Q = \left\{ \sum_i \delta_i X_i^{-\rho} \right\}^{-\frac{1}{\rho}}$$

- X_i = the quantity of input i
  Q = the quantity of output
  δ_i = share parameter for input i and $-1 < \rho < \infty$ is parameter
- *Assignment #1* - Show that the production function above exhibit constant returns to scale!
- With technical change terms:

$$Q = \frac{1}{A} \left\{ \sum_i \delta_i \left(\frac{X_i}{A_i} \right)^{-\rho} \right\}^{-\frac{1}{\rho}}$$

Elasticity of substitution

- Let's define σ:

$$\sigma = \frac{1}{1+\rho},\ 0 < \sigma < \infty$$

- We can substitute ρ with σ:

$$Q = \left\{\sum_i \delta_i X_i^{-\rho}\right\}^{-\frac{1}{\rho}} = \left\{\sum_i \delta_i X_i^{-\frac{1-\sigma}{\sigma}}\right\}^{-\frac{1}{\frac{1-\sigma}{\sigma}}}$$

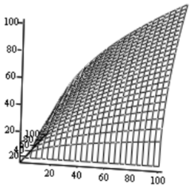
$$Q = \left\{\sum_i \delta_i X_i^{\frac{\sigma-1}{\sigma}}\right\}^{\frac{\sigma}{\sigma-1}}$$

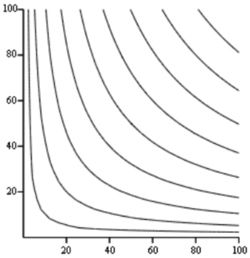
CES production function with varying parameters

- A two variables K, capital and L, labor:

$$Q = \left(\delta_K K^{\frac{\sigma - 1}{\sigma}} + \delta_L L^{\frac{\sigma - 1}{\sigma}} \right)^{\frac{\sigma}{\sigma - 1}}$$

- Surface and contour plot with $\delta_K = 0.5$, $\delta_L = 0.5$ and $\sigma = 0.9$





CES production function: Group exercise!

- Group exercise and discussion (15 minutes) - Using MathCAD file *CES_Curves.xmcd* in your laptop, try to change the following combination of parameters, and see how it affects the surface and counter plot of the function (note: you can rotate the surface plot by click and drag with your mouse)
- With this function: $Q = \left(\delta_K K^{\frac{\sigma-1}{\sigma}} + \delta_L L^{\frac{\sigma-1}{\sigma}}\right)^{\frac{\sigma}{\sigma-1}}$
  1. $\delta_K = 0.5, \sigma = 0.999$
  2. Bigger σ: $\delta_K = 0.5, \sigma = 2$
  3. Much bigger σ: $\delta_K = 0.5, \sigma = 100$
  4. Smaller σ: $\delta_K = 0.5, \sigma = 0.5$
  5. Much smaller σ: $\delta_K = 0.5, \sigma = 0.1$
  6. Bigger δ_K: $\delta_K = 0.75, \sigma = 0.999$
  7. Smaller δ_K: $\delta_K = 0.25, \sigma = 0.999$
- Discuss with your friends the effect of different combination of parameters.
- Try with $\sigma = 1$ and see what happens!

More about Sigma

- As you learned from the exercise, σ measure the curvature of an isoquant of the CES production function.
- σ is called the elasticity of substitution, it measures:
  - How easy it is for a producer to substitute between or among factors of production as responses to changing relative factor prices
  - How the change in the slope of the isoquant affect the factor intensity (ratio of two factors of production)
  - The percentage change in factor intensity as the effect of 1 percent change in factor prices ratio or marginal rate of technical substitution.
- $\sigma = 0.5$ means, when factor price ratio, let's say w/r increase by 1 percent, factor intensity will increase by 0.5.

Special cases of CES production function

- Cobb-Douglas Production function - A CES with $\sigma = 1$

$$Q = K^{\alpha} L^{\beta} \text{ or more generally: } Q = \prod_i X_i^{\alpha_i}$$

- Leontief production function - A CES with $\sigma = 0$

$$Q = \min\left(\frac{K}{a}, \frac{L}{b}\right) \text{ or more generally: } Q = \min\left(\frac{X_1}{A_1}, \ldots, \frac{X_n}{A_n}\right)$$

- Linear (perfect complement) production function - A CES with $\sigma = \infty$

$$Q = mK + nL, \text{ or more generally: } Q = \sum_i a_i X_i$$

Cost minimization

- A producer's problem is to minimize cost of production subject to a given production technology, or:

$$\min_{X_i} \sum_i W_i X_i \text{ s.t. } Q = \left\{ \sum_i \delta_i X_i^{-\rho} \right\}^{-\frac{1}{\rho}}$$

where W_i is the price of input i.

- To solve, we use the Lagrange function:

$$L = \sum_i W_i X_i + \lambda \left\{ Q - \left\{ \sum_i \delta_i X_i^{-\rho} \right\}^{-\frac{1}{\rho}} \right\}$$

- and find the first order conditions:

$$\frac{\partial L}{\partial X_i} = 0 \text{ for all } i = 1, \ldots, n \text{ and } \frac{\partial L}{\partial \lambda} = 0$$

Demand for or optimum quantity of factors of production

- From the first order conditions we can derive (See derivation in the handout, and ask if you get lost):

$$X_i = Q\delta_i^\sigma \left(\frac{W_i}{c}\right)^{-\sigma}$$

where:

$$c = \left(\sum \delta_i^\sigma W_i^{1-\sigma}\right)^{\frac{1}{1-\sigma}}$$

and σ is the elasticity of substitution, and $0 < \sigma < \infty$.

- c is the unit cost or average cost of production or:

$$c = \frac{C}{Q} = \frac{\sum_i W_i X_i}{Q}$$

See the derivation in the handout.

- Query: *why is the average cost a constant? shouldn't it be U-shaped?*

Demand for factors and unit cost

- You will see many of this in the equations of CGE models:

$$X_i = Q\delta_i^\sigma \left(\frac{W_i}{c}\right)^{-\sigma}$$

$$c = \left(\sum \delta_i^\sigma W_i^{1-\sigma}\right)^{\frac{1}{1-\sigma}}$$

- Make yourself familar (if not memorize), and think about what is the effect of changing Q and W_i into X_i.
  - X_i increase proportionally with Q, other things held constant: homotheticiy and constant returns to scale
  - X_i decrease with W_i (its own price), other things held constant
  - X_i increase with W_j (other input's price), other things held constant

Factor intensity and factor price ratio

- Factor intensity is the ratio of two optimal quantity of factors of production:

$$\frac{X_i}{X_j} = \frac{Q\delta_i^{\sigma}\left(\frac{W_i}{c}\right)^{-\sigma}}{Q\delta_j^{\sigma}\left(\frac{W_j}{c}\right)^{-\sigma}} = \left(\frac{\delta_i}{\delta_j}\frac{W_j}{W_i}\right)^{\sigma}$$

- Factor intensity, $\frac{X_i}{X_j}$, is a function of factor price ratio, $\frac{W_j}{W_i}$. This is a feature of a homothetic production function.
- This relationship explains the meaning of elasticity of substitution σ.
  - $\sigma = 1.5$ means that when factor price ratio $\frac{W_j}{W_i}$, increase by 1, factor intensity, $\frac{X_i}{X_j}$, will increase by 1.5.

Illustration with 2 factors production function

- With production function $Q = (\delta_K K^{-\rho} + \delta_L L^{-\rho})^{-\frac{1}{\rho}}$, and $\sigma = \frac{1}{1+\rho}$, the demand for factors of production are:

$$\begin{aligned} K &= Q\delta_K^{\sigma}\left(\frac{r}{c}\right)^{-\sigma} \\ L &= Q\delta_L^{\sigma}\left(\frac{w}{c}\right)^{-\sigma} \end{aligned}$$

where

$$c = \left(\delta_K^{\sigma} r^{1-\sigma} + \delta_L^{\sigma} w^{1-\sigma}\right)^{\frac{1}{1-\sigma}}$$

- For the purpose of visualization, the following equations define:
  - Isoquant: $K^Q(Q,L) = \left(\frac{Q^{\frac{\sigma-1}{\sigma}} - \delta_L L^{\frac{\sigma-1}{\sigma}}}{\delta_K}\right)^{-\frac{\sigma}{1-\sigma}}$
  - Isocost curve: $K^C(Q) = \frac{cQ-wL}{r}$
  - Factor intensity: $f = \left(\frac{\delta_K}{\delta_L}\frac{w}{r}\right)^{\sigma}$

Review

- The demand for inputs with CES technology, derived from cost minimization:

$$X_i = Q\delta_i^{\sigma}\left(\frac{W_i}{c}\right)^{-\sigma}$$

where

$$c = \left(\sum \delta_i^{\sigma} W_i^{1-\sigma}\right)^{\frac{1}{1-\sigma}}$$

The demand for inputs with Cobb-Douglas technology

- Producer's problem:

$$\min C = \sum_i W_i X_i \text{ subject to } Q = \prod_i X_i^{\alpha_i}$$

- Langrange equation:

$$L = \sum_i W_i X_i + \lambda \left(Y - \prod_i X_i^{\alpha_i} \right)$$

- First order conditions:

$$\frac{\partial L}{\partial X_i} = W_i - \alpha_i X_i^{\alpha_i - 1} \prod_{i \neq j} X_j^{\alpha_j} = 0$$

$$\frac{\partial L}{\partial \lambda} = Q - \prod_i X_i^{\alpha_i} = 0$$

The demand for inputs with Cobb-Douglas technology (cont)

- See derivation in the handout, the optimum X_i:

$$X_i = \frac{\alpha_i c Q}{W_i}$$

where

$$c = \prod_i \left(\frac{W_i}{\alpha_i}\right)^{\alpha_i}$$

is the unit cost of production.

The demand for inputs with Leontief technology

- Production function:

$$Q = \min \left\{ \frac{X_1}{A_1}, \ldots, \frac{X_n}{A_n} \right\}$$

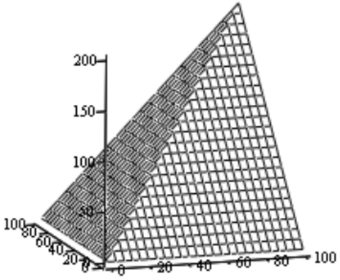
- A_i, where $0 < A_i < 1$ is called the Leontief technical coefficient.
- The optimum (cost-minimizing) or the demand for inputs:

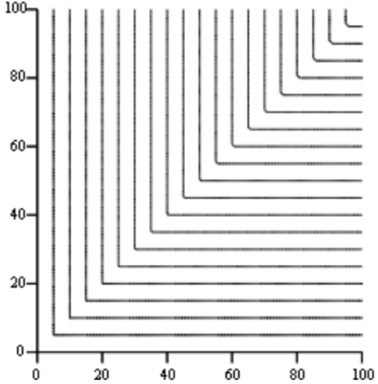
$$X_i = A_i Q$$

It is proportional to the quantity of output, independent of input prices.

Leontief Production Function - Curvature

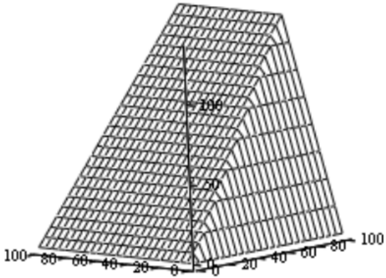
$Q = \min\left\{\frac{K}{A_K}, \frac{L}{A_L}\right\}$, with $A_K = A_L = 0.5$

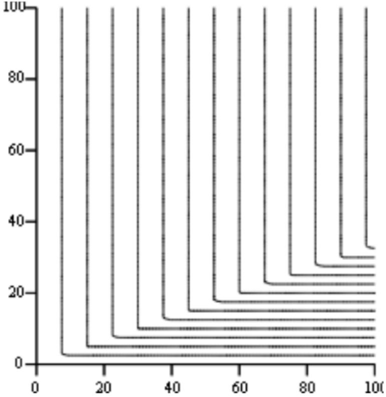




Leontief Production Function - Curvature

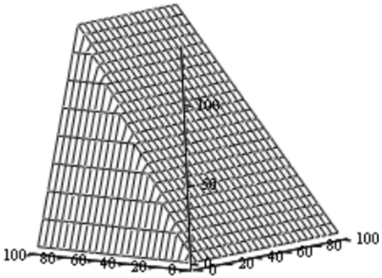
$Q = \min\left\{\frac{K}{A_K}, \frac{L}{A_L}\right\}$, with $A_K = 0.25, A_L = 0.75$

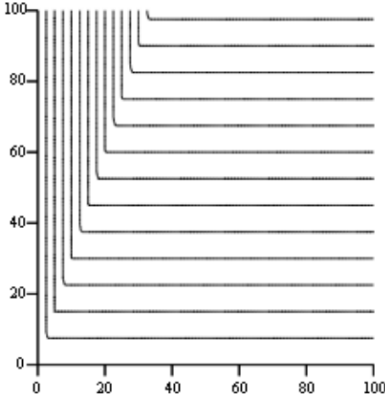




Leontief Production Function - Curvature

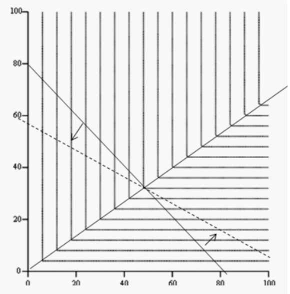
$$Q = \min\left\{\frac{K}{A_K}, \frac{L}{A_L}\right\}, \text{ with } A_K = 0.75, A_L = 0.25$$





Leontief production function is input-price independent

- $Q = \min\left\{\frac{K}{A_K}, \frac{L}{A_L}\right\}$, with $A_K = 0.4, A_L = 0.6$



- Factor intensity $f = \frac{K}{L} = \frac{0.4}{0.6} = \frac{2}{3}$ independent of prices, or $\sigma = 0$
- Cost minimizing inputs: $K = 0.4Q, L = 0.6Q$

Nested Production function

- Suppose we have a 3-inputs production function:

$$Q = f(U, S, K)$$

  where: Q = Output, U = Unskilled labor
  S = Skilled labor, and K = Capital

- The producer's standard problem is to:

$$\min w_U U + w_S S + rK \text{ subject to } Q = f(U, S, K)$$

  where w_U, w_S and r are unskilled-wage, skilled-wage, and return to capital respectively.

- The standard procedure for optimization can be used to find the cost-minimizing U, S and K:

$$L = w_U U + w_S S + rK + \lambda(Q - f(U, S, K))$$

Nested production function

- Now, suppose the production function looks like this:

$$Q = \left(\delta_L \left(\delta_U U^{-\rho_L} + \delta_S S^{-\rho}\right)^{\frac{\rho}{\rho_L}} + \delta_K^{-\rho}\right)^{-\frac{1}{\rho}}$$

- This is an example of the *nested*, or *separable* production function, because we can re-write:

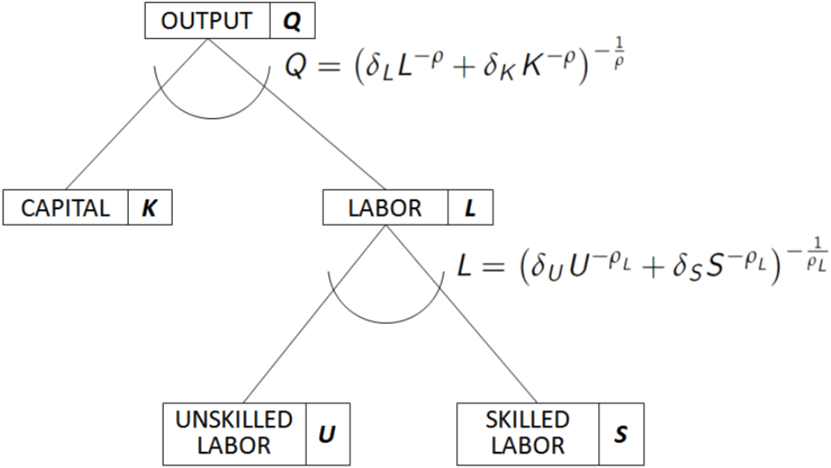
$$Q = f\left(L, K\right) = \left(\delta_L L^{-\rho} + \delta_K K^{-\rho}\right)^{-\frac{1}{\rho}}$$

where

$$L = g\left(U, S\right) = \left(\delta_U U^{-\rho_L} + \delta_S S^{-\rho_L}\right)^{-\frac{1}{\rho_L}}$$

- L is often called a composite variable because it is a composite or CES aggregation of U and S.
- Both $f\left(\cdot\right)$ and $g\left(\cdot\right)$ have CES form but with different parameter, thus different elasticity of substitution.

Diagram of a nested production function



Cost minimization with nested production function

- How to solve the minimization problem with nested production function like this?

$$\min_{U,S,K} w_U U + w_S S + rK \text{ s.t. } Q = \left(\delta_L \left(\delta_U U^{-\rho_L} + \delta_S S^{-\rho_L}\right)^{\frac{\rho}{\rho_L}} + \delta_K^{-\rho}\right)^{-\frac{1}{\rho}}$$

- We can do it in steps, by dividing the problem into two problems:
- Problem #1: Find optimum combination of U and S for a given L (composite labor):

$$\min_{U,S} w_U U + w_S S \text{ s.t } L = \left(\delta_U U^{-\rho_L} + \delta_S S^{-\rho_L}\right)^{-\frac{1}{\rho_L}}$$

- Problem #2: Find optimum combination of K and L for a given Q:

$$\min_{K,L} wL + rK \text{ s.t. } Q = \left(\delta_L L^{-\rho} + \delta_K K^{-\rho}\right)^{-\frac{1}{\rho}}$$

where w is the price of composite labor L.

Solution to problem #1

- Problem:

$$\min_{U,S} w_U U + w_S S \text{ s.t } L = \left(\delta_U U^{-\rho_L} + \delta_S S_L^{-\rho}\right)^{-\frac{1}{\rho_L}}$$

- We can apply the standard formula:

$$X_j = Q\delta_j^{\frac{1}{1+\rho}} \left(\frac{W_j}{c}\right)^{-\frac{1}{1+\rho}} \text{ and } c = \left(\sum \delta_i^{\frac{1}{1+\rho}} W_i^{\frac{\rho}{1+\rho}}\right)^{\frac{1+\rho}{\rho}} \text{ to find:}$$

$$U = L \cdot \delta_U^{\frac{1}{1+\rho_L}} \left(\frac{w_U}{w}\right)^{-\frac{1}{1+\rho_L}} \quad (1)$$

$$S = L \cdot \delta_S^{\frac{1}{1+\rho}} \left(\frac{w_S}{w}\right)^{-\frac{1}{1-\rho_L}} \quad (2)$$

$$w = \left(\delta_U^{\frac{1}{1+\rho_L}} w_U^{\frac{\rho_L}{1+\rho_L}} + \delta_S^{\frac{1}{1+\rho_L}} w_S^{\frac{\rho_L}{1+\rho_L}}\right)^{\frac{1+\rho_L}{\rho_L}} \quad (3)$$

Solution to problem #2

- Problem:

$$\min_{L,K} wL + rK \text{ s.t } Q = \left(\delta_L L^{-\rho} + \delta_K K^{-\rho}\right)^{-\frac{1}{\rho}}$$

- We can apply the standard formula:

$$X_j = Q\delta_j^{\frac{1}{1+\rho}} \left(\frac{W_j}{c}\right)^{-\frac{1}{1+\rho}} \text{ and } c = \left(\sum \delta_i^{\frac{1}{1+\rho}} W_i^{\frac{\rho}{1+\rho}}\right)^{\frac{1+\rho}{\rho}} \text{ to find:}$$

$$L = Q \cdot \delta_L^{\frac{1}{1+\rho}} \left(\frac{w}{c}\right)^{-\frac{1}{1+\rho}} \quad (4)$$

$$K = Q \cdot \delta_K^{\frac{1}{1+\rho}} \left(\frac{r}{w}\right)^{-\frac{1}{1-\rho}} \quad (5)$$

$$c = \left(\delta_L^{\frac{1}{1+\rho}} w^{\frac{\rho}{1+\rho}} + \delta_K^{\frac{1}{1+\rho}} r^{\frac{\rho}{1+\rho}}\right)^{\frac{1+\rho}{\rho}} \quad (6)$$

Complete solution

Equation 1 to 9 define the solution of the problems: the optimum quantity of U, S, K for a given Q, w_U, w_S, r:

1. $w = \left(\delta_U^{\frac{1}{1+\rho_L}} w_U^{\frac{\rho_L}{1+\rho_L}} + \delta_S^{\frac{1}{1+\rho_L}} w_S^{\frac{\rho_L}{1+\rho_L}} \right)^{\frac{1+\rho_L}{\rho_L}}$
2. $c = \left(\delta_L^{\frac{1}{1+\rho}} w^{\frac{\rho}{1+\rho}} + \delta_K^{\frac{1}{1+\rho}} r^{\frac{\rho}{1+\rho}} \right)^{\frac{1+\rho}{\rho}}$
3. $K = Q \cdot \delta_K^{\frac{1}{1+\rho}} \left(\frac{r}{w}\right)^{-\frac{1}{1-\rho}}$
4. $L = Q \cdot \delta_L^{\frac{1}{1+\rho}} \left(\frac{w}{c}\right)^{-\frac{1}{1+\rho}}$
5. $U = L \cdot \delta_U^{\frac{1}{1+\rho_L}} \left(\frac{w_U}{w}\right)^{-\frac{1}{1+\rho_L}}$
6. $S = L \cdot \delta_S^{\frac{1}{1+\rho}} \left(\frac{w_S}{w}\right)^{-\frac{1}{1-\rho_L}}$

Solution to problem #2

- Problem:

$$\min_{L,K} wL + rK \text{ s.t } Q = \left(\delta_L L^{-\rho} + \delta_K K^{-\rho}\right)^{-\frac{1}{\rho}}$$

- We can apply the standard formula:

$$X_j = Q\delta_j^{\frac{1}{1+\rho}} \left(\frac{W_j}{c}\right)^{-\frac{1}{1+\rho}} \text{ and } c = \left(\sum \delta_i^{\frac{1}{1+\rho}} W_i^{\frac{\rho}{1+\rho}}\right)^{\frac{1+\rho}{\rho}} \text{ to find:}$$

$$L = Q \cdot \delta_L^{\frac{1}{1+\rho}} \left(\frac{w}{c}\right)^{-\frac{1}{1+\rho}} \quad (7)$$

$$K = Q \cdot \delta_K^{\frac{1}{1+\rho}} \left(\frac{r}{w}\right)^{-\frac{1}{1-\rho}} \quad (8)$$

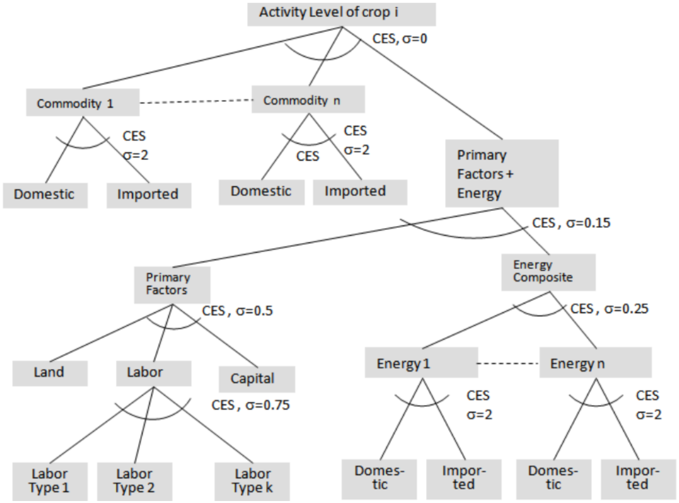
$$c = \left(\delta_L^{\frac{1}{1+\rho}} w^{\frac{\rho}{1+\rho}} + \delta_K^{\frac{1}{1+\rho}} r^{\frac{\rho}{1+\rho}}\right)^{\frac{1+\rho}{\rho}} \quad (9)$$

Complete solution

Equation 1 to 9 define the solution of the problems: the optimum quantity of U, S, K for a given Q, w_U, w_S, r:

1. $$w = \left(\delta_U^{\frac{1}{1+\rho_L}} w_U^{\frac{\rho_L}{1+\rho_L}} + \delta_S^{\frac{1}{1+\rho_L}} w_S^{\frac{\rho_L}{1+\rho_L}} \right)^{\frac{1+\rho_L}{\rho_L}}$$
2. $$c = \left(\delta_L^{\frac{1}{1+\rho}} w^{\frac{\rho}{1+\rho}} + \delta_K^{\frac{1}{1+\rho}} r^{\frac{\rho}{1+\rho}} \right)^{\frac{1+\rho}{\rho}}$$
3. $$K = Q \cdot \delta_K^{\frac{1}{1+\rho}} \left(\frac{r}{w}\right)^{-\frac{1}{1-\rho}}$$
4. $$L = Q \cdot \delta_L^{\frac{1}{1+\rho}} \left(\frac{w}{c}\right)^{-\frac{1}{1+\rho}}$$
5. $$U = L \cdot \delta_U^{\frac{1}{1+\rho_L}} \left(\frac{w_U}{w}\right)^{-\frac{1}{1+\rho_L}}$$
6. $$S = L \cdot \delta_S^{\frac{1}{1+\rho}} \left(\frac{w_S}{w}\right)^{-\frac{1}{1-\rho_L}}$$

Nesting can be complex, but standard principle is the same



Assignment #2

Write down the equations that define the solution to the following problem:

$$\text{Minimize cost: } C = w_U U + w_S S + p_N N + rK + p_E E + p_D X_D + p_M X_M$$

$$\text{Subject to: } Q = \min\left\{ \frac{f_\rho\left(f_{\rho US}\left(U, S\right), f_{\rho KE}\left(K, E\right), N\right)}{a_1}, \frac{f_{\rho DM}\left(X_D, X_M\right)}{a_2} \right\}$$

Where:

U = unskilled labor	w_U = unskilled wage
S = skilled labor	w_S = skilled wage
N = land	p_N = return to land
E = energy	p_E = price of energy
K = capita	r = return to capital
X_D = domestic raw material	p_D = price of X_D
X_M = imported raw material	p_M = price of X_M

$f_\rho, f_{\rho US}, f_{\rho KE}, f_{\rho DM}$ is a CES function with parameter $\rho, \rho_{US}, \rho_{KE}, \rho_{DM}$ respectively.
a_1 and a_2 is the Leontief technical coefficient.

Assignment #2: Diagram